

	Name
	Class
	R. No.

	• Physics • +1 CBSE/ IITJEE
	• Test 1 (August 3, 2010) • Mathematical Tools(Derivatives)
	• Maximum Marks 100 • Time 40 minutes

Instructions

1. The test contains four questions.
2. Marks will be awarded for partially correct answers also.
3. Roughly 24 seconds are allotted for each mark.

Q1: Find the derivatives of the following functions with respect to x

a) $f(x) = (ax - a)^m (bx - b)^n (cx - c)^p$

8 Marks

b) $f(x) = \frac{(ax - \alpha)(bx - \beta)(cx - \gamma)}{(ax + \alpha)(bx + \beta)(cx + \gamma)}$

8 Marks

Space for Rough Work

c) $f(x) = \sin(x^{\log 3})$

8 Marks

d) $f(x) = \log(\cos(e^{3x}))$

8 Marks

Space for Rough Work

e) $f(x) = \sin^{-1}(\cos x)$

8 Marks

Space for Rough Work

Q2: Find $\frac{dy}{dx}$ if $x^3 + y^2x^2 + xy^4 + 2x + 15y + 53 = 0$

20 Marks

Q3: Q: Find $\frac{dy}{dx}$ if $x = a(t - \sin t)$ and $y = a(1 - \cos t)$

20 Marks

Q4: Find $\frac{d^3y}{dx^3}$ if x and y are expressed parametrically as $x = e^{-t}$ and $y = t^3$

20 Marks